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Low-loss waveguides in a single-crystal lithium niobate thin film

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We report low-loss channel waveguides in a single-crystal LiNbO_3 thin film achieved using the annealed proton exchange process. The simulation indicated that the mode size of the α phase channel waveguide could be as small as $1.2 \mu\text{m}^2$. Waveguides with several different widths were fabricated, and the $4 \mu\text{m}$ -wide channel waveguide exhibited a mode size of $4.6 \mu\text{m}^2$. Its propagation loss was accurately evaluated to be as low as 0.6 dB/cm at $1.55 \mu\text{m}$. The single-crystal lattice structure in the LiNbO_3 thin film was preserved by a moderate annealed proton exchange process (5 min of proton exchange at 200°C , followed by 3 h annealing at 350°C), as revealed by measuring the extraordinary refractive index change and x ray rocking curve. A longer proton exchange time followed by stronger annealing would destroy the crystal structure and induce a high loss in the channel waveguides. © 2015 Optical Society of America

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A single-crystal lithium niobate (LN) thin film (lithium niobate on insulator, LNOI) is an ideal platform for integrated optics due to its high index contrast, which enables the tight confinement of light [1]. By using crystal ion slicing and wafer bonding methods, single-crystal LN thin films that possess excellent optical, nonlinear, and electro-optical properties as bulk materials have been reported [2,3], and various photonic devices have been fabricated using them, such as photonic wires [4], photonic crystal slabs [5], electro-optic modulators [6], and micro-disk resonators [7]. The channel waveguide is a fundamental structure for integrated optics. For channel waveguides in LNOIs, light can be confined to a much smaller size than in bulk LN [8], and strong electro-optical and nonlinear optical effects are expected. However, the waveguides in the LN thin films currently reported have a loss around 10 dB/cm [4,8], which is too large for applications such as nonlinear wavelength conversions, Mach-Zehnder modulators, and waveguide lasers.

Proton exchange (PE) is an efficient and mature technology used to produce low-loss optical devices in bulk LN materials [9]. Among all seven PE phases in LN, the α phase waveguide has the smallest propagation loss, and the waveguide region has similar electro-optic and nonlinear coefficients to that of bulk LN due to the analogous lattice structure [10,11]. A low-loss annealed PE (APE) waveguide with a compact mode size in a LNOI is thus needed to enhance the performance and integration level of the integrated optical devices. In this Letter, a low-loss (0.6 dB/cm) and small mode size ($4.6 \mu\text{m}^2$) APE channel waveguide was fabricated. The characteristics of the guided modes were given by a full-vectorial finite-difference method [12]. The ratio of the light energy confined in the APE region to the total energy exceeded 50% when the width was larger than 1.9 and $4.5 \mu\text{m}$ for the fundamental mode (TM_0) and the first higher-order mode (TM_1), respectively. The mode size could reach as small as 1.2 for a $2.5 \mu\text{m}$ wide channel waveguide in a $0.5 \mu\text{m}$ -thick LN film. The PE channel waveguides in the LNOI had a zero cut-off width of the channels for TM_0 [13], while in the bulk LN, TM_0 could exist only if the width exceeded a critical value. A much smaller mode size was obtained compared with the APE in the bulk LN. The phase change during the APE process was characterized by x-ray diffraction and a prism coupling method.

Compared with PE in bulk LN, proton diffusion in LNOI was more complex, because the protons might have different diffusion rates in SiO_2 and LN. According to a previous study about the proton diffusion mechanism in amorphous SiO_2 [14], the activation energy for protons in SiO_2 (below 1 eV , depending on the different mechanisms and experiments) was smaller than that in LN (1.21 eV) [15]. So the amorphous SiO_2 layer probably could not prevent the protons from diffusing to a deeper place. The refractive index change in SiO_2 by the diffused protons was ignored because the index difference between LN and SiO_2 was large. Therefore, in the simulation, the model was based on the refractive index distribution of an APE bulk LN. The top 560 nm thin film was untouched, while the bottom part was replaced by a SiO_2 layer. The parameters related to the APE process, such as the diffusion coefficients, were determined according to the work in [15]. The PE time and anneal time were 5 min and 3 h , respectively (this condition was used to fabricate the channel waveguides, as discussed below).

A simulated refractive index distribution in LNOI according to the above APE condition is shown in Fig. 1(a). Since the channel waveguide was formed by adding a high-index region (APE) into the LN film planar waveguide, this type of waveguide was a strip-loaded waveguide, which had several unique characteristics compared to the conventional APE waveguides in bulk LN [13]. For example, in LNOI, the APE region (the rectangular region with width W , which is equal to the width of the initial PE region and height H , which is the thickness of the LN film) with an arbitrary cross-section would guide the light. It exhibited a zero cut-off width for TM_0 since the structure was symmetrical in the lateral direction [13]. But in contrast, in bulk LN, the light could be guided only if the dimension of the waveguide core was large enough. Moreover, the mode size in LNOI could reach a much smaller value because the mechanism of the wave guiding condition was that the effective refractive indices of the guided modes were larger than that of the nearby planar waveguide (about 1.82 at 1.55 μm for a 0.56 μm -thick LN film) instead of the surrounding LN refractive index ($n_e = 2.138$ at 1.55 μm) [8]. All of these properties made the APE waveguide or integrated optical devices based on the APE waveguide in LNOI very promising.

As shown in Fig. 1(b), the ratio of the light energy of TM_0 and TM_1 confined in the APE region to the total energy increased with an increase in W . This ratio exceeded 50% when W was larger than 1.9 and 4.5 μm for TM_0 and TM_1 , respectively.

A small mode size was needed to enhance the nonlinear optical process. The relationships between the mode sizes (the product of $1/e$ intensity in the horizontal and vertical directions) and W are given in Fig. 2 at several LN film thicknesses. When W was small, the mode sizes decreased as W increased. Then, they reached a minimal value around $W = 2.5 \mu\text{m}$, except for the 0.3 μm -thick waveguide, for which the minimal value was around 3 μm . If W continued to increase, the mode sizes became larger along with the expanded APE region. The dependence of the mode sizes on H was similar to that on W . The mode sizes decreased as H increased at first, and then reached a minimal value around $H = 0.5 \mu\text{m}$. If H was larger than 0.5 μm , the mode sizes would increase with H . The

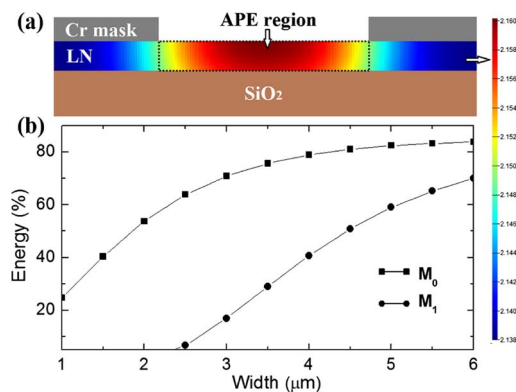


Fig. 1. (a) Cross section of the APE waveguide in LNOI (Cr mask was shown to illustrate the width of the initial PE region). The scale bar on the right side gives the refractive index distribution represented by different colors in the LN film at 1.55 μm . (b) Light energy of TM_0 and TM_1 confined in the APE region (surrounded by the black dotted line in the top picture) as a function of the initial width of the PE region.

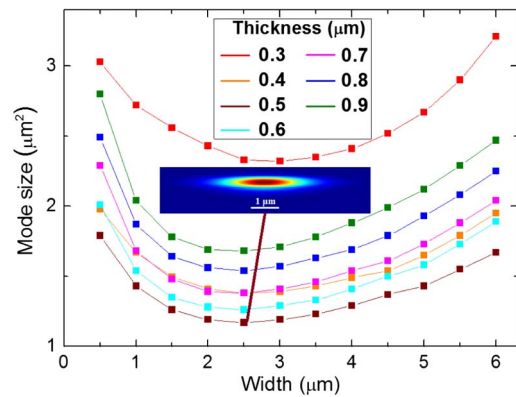


Fig. 2. Relationship between the mode sizes and W for various LN film thicknesses in the simulation. The smallest value, about 1.2 μm^2 at 1.55 μm , was reached when $W = 2.5 \mu\text{m}$. Inset: simulated near-field intensity distribution of the TM_0 of a 2.5 μm -wide channel waveguide in a 0.5 μm -thick LN film.

smallest size, 1.2 μm^2 , was reached around $W = 2.5 \mu\text{m}$ for a 0.5 μm -thick LN film. The near-field intensity distribution of this smallest mode is shown in the inset of Fig. 2. This value was much smaller than the APE waveguide with the α phase in the bulk LN [10].

In our experiment, several identical 560 nm-thick z-cut single-crystal LNOIs were prepared by the crystal ion slicing and wafer bonding methods. The PE processes were performed by immersing the samples in benzoic acid under different PE and annealing conditions. First, 15 min of PE at 200°C, which has been widely used in bulk LN, was used to explore the possibility of making α phase waveguides in the LN film. The high resolution x ray diffraction (HRXRD) rocking curves spectrum is shown in Fig. 3(a); it exhibits complicated PE-induced phases, including β_3 and mixed phases. The different phases were identified by previous research [16]. After an hour-long anneal at 350°C, the κ_2 and α phases appeared [see Fig. 3(b)]. The mean extraordinary refractive index change (Δn_e) of the LN film at 633 nm, which was measured by the prism coupling method, was 0.0593. This value was much higher than the criterion of the α phase index change (<0.02) [17]. To get the pure α phase, the anneal time was extended (the temperature was fixed at 350°C). However, Δn_e changed slightly even after additional 6-hour-long anneal. Then, another sample with the same PE condition was annealed at 400°C. After a 1-hour anneal, Δn_e was still high, but after additional 1-hour anneal, Δn_e decreased to as low as 0.013, which met criterion of the α phase index change. From its HRXRD spectrum [see Fig. 3(c)], however, it can be seen that the LN film diffraction peak almost disappeared, indicating that the crystal lattice in the LN film was probably destroyed by the APE process. To evaluate the performance of the APE waveguide, channel waveguides were fabricated with this APE condition and a very low intensity of transmitted light was found, indicating a high propagation loss. This phenomenon was consistent with the HRXRD results [see Fig. 3(c)]. Therefore, the typical APE process used in bulk LN made it difficult to get the low-loss α phase in LNOI. The PE condition was improved by decreasing the PE time to 5 min to relieve the impact of the diffused protons on the initial crystal structure, followed by sufficiently long time spent annealing

Table 1. Effective Refractive Indices of the TM Modes at 633 nm (PE at 200°C for 5 min, Annealed at 350°C 3 for Hours)

	Mode0	Mode1	Mode2	Refractive Index
Before PE	2.1452	1.9584	1.6410	2.2053
After APE	2.1577	1.9672	1.6444	2.2189

(3 h at 350°C) to get the pure α phase. The TM guided modes are shown in Table 1, and the value of Δn_e is 0.0136 (< 0.02). All of the effective refractive indices of the TM modes increased after the APE process. The HRXRD spectrum in Fig. 3(d) shows a good single-crystal lattice in the LN film, although the α phase diffraction peak was too weak to observe, probably due to the relative low proton concentration in such a short PE time. But phase change could possibly happen (the possible phase was α phase, since the other phases were much more obvious in x ray diffraction and they did not appear). Using this PE condition, the low-loss channel waveguides shown below were fabricated. The full width at half maximum of the LN film diffraction peak in Fig. 3(d) was 86° , which was smaller than 101° in Fig. 3(b), indicating a good preservation of the crystal lattice in the PE condition in Fig. 3(d).

Another 560 nm-thick z-cut LNOI was used to fabricate the channel waveguides. A 60 nm-thick chromium (Cr) film was deposited on the sample surface by an electron beam evaporator. Then, the 1–7 μm wide-open strips in the Cr film were patterned by photolithography and wet etching. After the photoresist was removed by acetone, the PE process was performed at 200°C for 5 min. The residual benzoic acid and the Cr mask were removed by alcohol and Cr etchant, respectively, and then the anneal was carried out at 350°C for 3 h [the APE condition was the same as that in Fig. 4(d)]. Finally, the two end faces of the sample were polished, and the sample was cleaned for measurement. The polishing was performed after the anneal because the diffusion of the protons would become inhomogeneous near the end face during the anneal. The subsequent polishing would remove this inhomogeneous part. For the channel waveguide case, there must be some lateral proton

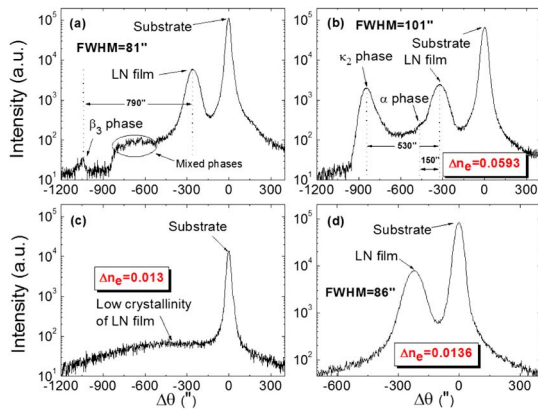


Fig. 3. HRXRD rocking curves spectra (logarithm of intensity) for (0012) reflection of APE LNOI under conditions of (a) PE at 200°C for 15 min without anneal, (b) PE at 200°C for 15 min, anneal at 350°C for 1 h, (c) PE at 200°C for 15 min, anneal at 400°C for 2 h, and (d) PE at 200°C for 5 min, anneal at 350°C for 3 h. The different phases with specific diffraction angles (deviations from the LN film peak) are labeled.

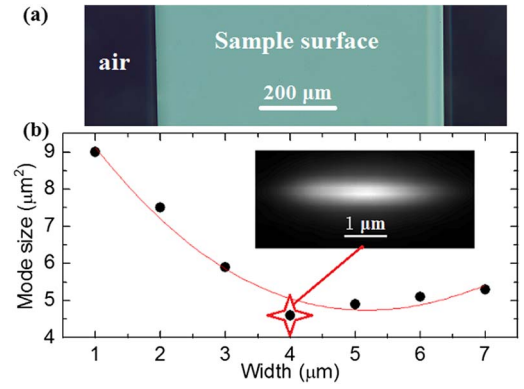


Fig. 4. (a) Optical microscope image (top view) of the sample surface. (b) Measured mode sizes of the 1–7 μm wide channel waveguides. Inset: measured near-field intensity distribution of the 4 μm -wide channel waveguide, which had the smallest mode size, $4.6 \mu\text{m}^2$.

diffusion during the anneal, and the proton concentration should be lower than the planar waveguide case, leading to a smaller refractive index change. After polishing the end faces, the final length of the sample was 0.8 mm. The top view of the sample surface is shown in Fig. 4(a). To optically characterize the channel waveguides, a polarization maintaining fiber, which had a lensed tip to improve the mode matching between the waveguide and the fiber, was used to transmit a $1.55 \mu\text{m}$ TM-polarized light from a tunable laser and couple it into the waveguide. The transmitted light was collected by a $40\times/0.65$ objective with an antireflection coating, and was detected by a beam profiler. In Fig. 4, the measured mode sizes are summarized. The mode size increased along with the expansion of the waveguide width. On the other hand, the effective refractive index of the guided mode also increased with the width, leading to a stronger confinement of the light. So the relationship between the mode size and width was a competitive result of these two factors, exhibiting a minimal value. This trend is similar to the simulation (see Fig. 2), although the minimal mode size positions did not coincide very well. This was probably due to the different diffusion coefficients in the simulation and in the actual case. The 4 μm -wide waveguide had the most compact mode size, $4.6 \mu\text{m} \times 1 \mu\text{m} = 4.6 \mu\text{m}^2$, and its near-field intensity distribution is shown in the inset of Fig. 4. The simulated waveguide had a mode size of $3.3 \mu\text{m} \times 0.4 \mu\text{m} = 1.3 \mu\text{m}^2$. The difference was mainly due to the fact that the vertical size of the guided mode ($0.4 \mu\text{m}$) was much smaller than the objective resolution ($\sim 1.5 \mu\text{m}$). The real mode profile and dimension in the vertical direction could not be completely reflected, leading to the larger mode size.

Figure 5 shows the transmission spectra of the 4 μm -wide channel waveguide. The transmitted light was recorded by a germanium photodiode with a resolution of 1 nW. Based on the Fabry–Perot method, the propagation loss α was evaluated by the following Eq. [18]:

$$\alpha = \frac{4.34}{L} (\ln R - \ln \tilde{R}) \quad \text{where } \tilde{R} = \frac{1}{K} \left(1 - \sqrt{1 - K^2} \right)$$

$$\text{and } K = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}.$$

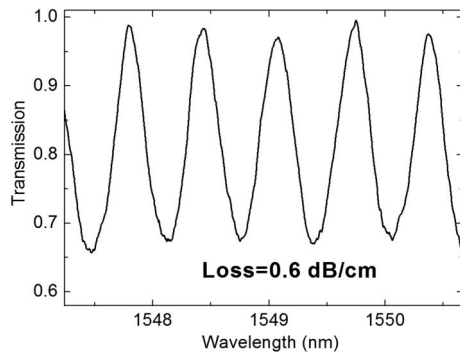


Fig. 5. Normalized transmission of the 4 μm -wide channel waveguide as a function of the wavelength.

I_{max} and I_{min} are the maximum and minimum intensity of the transmitted light extracted from Fig. 5. The value of the loss was closely related to the end face reflectivity of the waveguides. Therefore, a finite difference time domain (FDTD) solver [12] was used to calculate the end face reflectivity with different values of W . The end face reflectivity exhibited a maximal value when W was 4.5 μm , as shown in Fig. 6(a). This is due to the fact that, during the anneal, the lateral diffusion of the protons would induce a decrease in the refractive index. The smaller the width was, the more obvious the decrease was. So, as the width increased, the refractive index became larger, leading to an increased end face reflectivity. For a large width, the increase of the refractive index slowed down (lateral diffusion had a limited effect on the refractive index change), and the end face reflectivity (TM mode) would decrease as the waveguide core size increased [4]. In the experiments, the polished end faces were not always exactly perpendicular to the channels. So the end face reflectivity at different angles was calculated. The end face reflectivity decreased dramatically with the angle, as shown in Fig. 6(b). This resulted in a big error in the exact evaluation of the propagation loss. So the end faces of the channel waveguides were etched to be perpendicular to the channels by focused ion beam milling. The propagation loss of the 4 μm -wide channel, which had the smallest mode size, was as low as 0.6 dB/cm. Such a low propagation loss indicated that the crystal ion slicing process induced minor loss in the waveguide. In addition, compared with the Ar^+ milling waveguides [4], the APE method produced waveguides by modifying the crystal phase, avoiding the rough etched walls during Ar^+ milling. Thus, this method greatly reducing the loss.

In conclusion, using the APE process, the mode size of an α phase channel waveguide could be as small as 1.2 μm^2 for a 560 nm-thick LN film. HRXRD rocking curves and a prism coupling method were used to determine the most suitable condition to realize the α phase APE waveguide in LNOI. A long PE time with a strong anneal would destroy the lattice structure in the LN film and induce a high loss in the channel waveguides. A short PE time (5 min at 200°C) followed by a sufficient anneal (3 h at 350°C) was used, and the mean extraordinary refractive index change of the LN film after the APE was measured to be 0.0136 at 633 nm, which met the α phase condition ($\Delta n_e < 0.02$). Channel waveguides with different widths were fabricated, and a 4 μm -wide waveguide

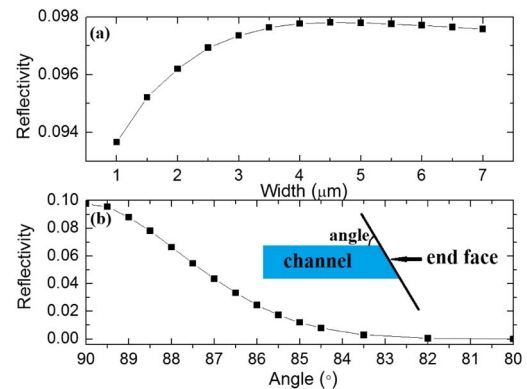


Fig. 6. Dependence of the end face reflectivity on (a) the waveguide width; (b) the angle between the end face and the channels. Inset: top view of the channel waveguide showing the angle between the end face and the channel direction.

exhibited the smallest mode size of 4.6 μm^2 at 1.55 μm . As revealed by the FDTD calculation of the end face reflectivity with different widths and slope angles, the end faces of the waveguides were etched to be exactly perpendicular to the channels. In this way, the propagation loss was accurately evaluated to be as low as 0.6 dB/cm at 1.55 μm by Fabry–Perot method. Based on such low-loss waveguides with small mode sizes in LNOI, high performance integrated optical devices, such as micro-ring, modulator, and nonlinear devices could successfully be fabricated.

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